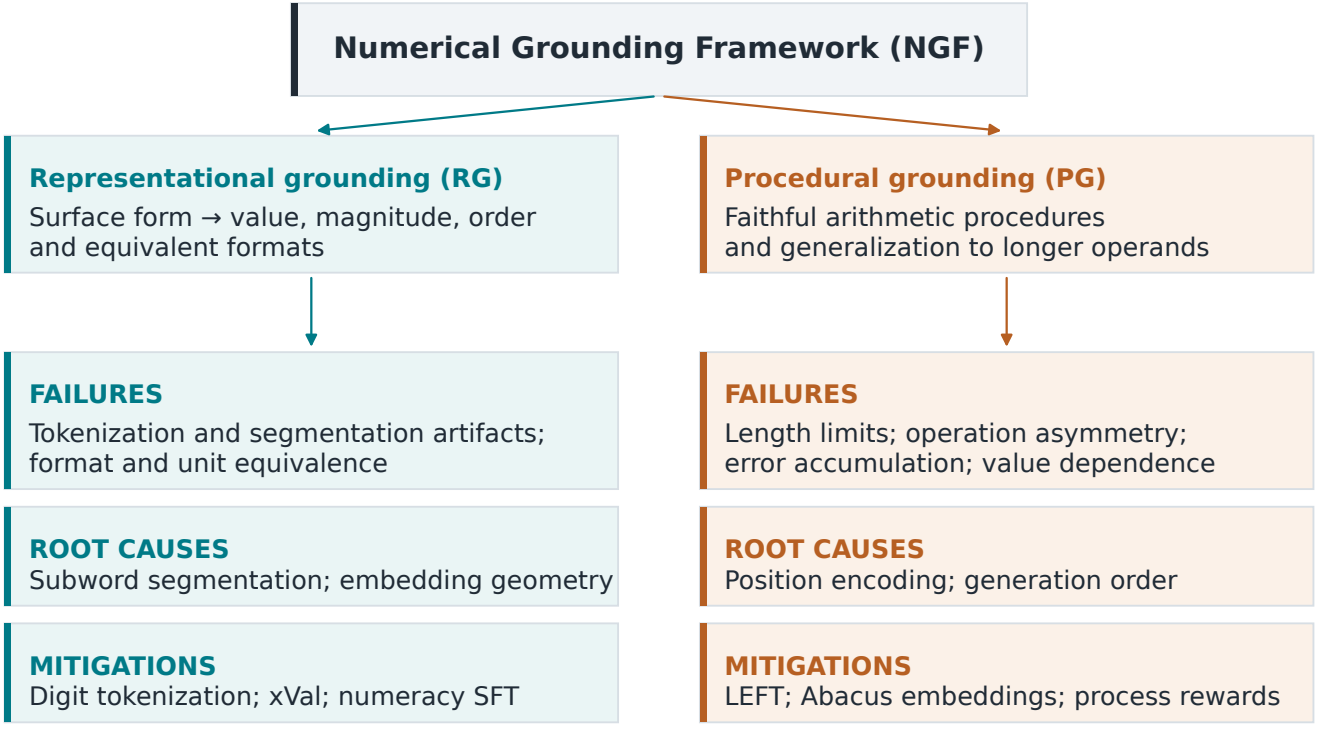
Numerical Grounding Framework (NGF)
Representational grounding (RG)
Surface form → value, magnitude, order and equivalent formats
Procedural grounding (PG)
Faithful arithmetic procedures and generalization to longer operands
FAILURES
Tokenization and segmentation artifacts; format and unit equivalence
FAILURES
Length limits; operation asymmetry; error accumulation; value dependence
ROOT CAUSES
Subword segmentation; embedding geometry
ROOT CAUSES
Position encoding; generation order
MITIGATIONS
Digit tokenization; xVal; numeracy SFT
MITIGATIONS
LEFT; Abacus embeddings; process rewards
Outside NGF (upstream): operand selection, i.e. using the wrong quantities
Robust numeracy requires both RG and PG